

DC MOTOR SOFT STARTER

Using Arduino Uno Board

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Motor starter reduces the load, torque and current surge of a motor during startup. On starting, the motor takes more than five times the normal running current. This overheats the motor's armature winding and creates a sudden voltage dip in the power supply, which can be avoided by using a motor starter.

There are many types of motor starters. Here we describe an electronic DC motor soft-starter using Arduino Uno board. This circuit controls both soft-start and soft-stop timings through pulse-width modulation (PWM).

Circuit and working

Circuit diagram of the DC motor soft-starter is shown in Fig. 1. In addition to Arduino Uno board (Board1), it uses PIC817 optocoupler (IC1), p-channel IRF9530 MOSFET (T1), 1N4007 rectifier diode (D1), 12V DC motor (M1) for testing, bi-colour LED (LED1) and a few other components.

Arduino Uno board is an important part of the circuit that generates PWM signals. Pushbuttons (S1 and S2) are used to soft-start and soft-stop the motor. The bi-colour LED indicates whether the motor is in soft-start or soft-stop mode.

Soft-start. When the circuit is switched on, the motor is at rest and LED1 off. When pushbutton S1 is pressed momentarily, the voltage across the motor increases gradu-

ally and attains maximum voltage after the predetermined soft-start timing. So the motor first starts at a slow speed. In this state, LED1 emits green light. After the soft-start time, i.e., in running condition, LED1 emits orange light, indicating that the motor is running at the rated speed.

Soft-stop. If pushbutton S2 is pressed momentarily while the motor is running at the rated speed, the voltage across the motor decreases gradually and becomes zero after the soft-stop time. So the motor speed also decreases gradually and it finally stops after the soft-stop time. In this state, LED1 emits red colour. After the soft-stop time, LED1 doesn't glow, indicating that the motor is at rest.

Arduino Uno is programmed such that if switch S1 or S2 is

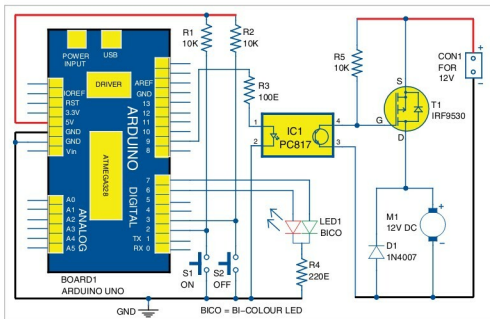


Fig. 1: Circuit diagram of DC motor soft-starter

PARTS LIST

Semiconductors:

- Board1 - Arduino Uno board
- IC1 - PIC817 optocoupler
- T1 - IRF9530 p-channel MOSFET
- D1 - 1N4007 rectifier diode
- LED1 - 5mm bi-colour LED

Resistors (all 1/4-watt, ±5% carbon):

- R1, R2, R5 - 10-kilo-ohm
- R3 - 100-ohm
- R4 - 220-ohm

Miscellaneous:

- CON1 - 2-pin terminal connector
- CON2 - 8-pin connector
- M1 - 12V DC motor for testing
- S1, S2 - Male-to-male jumper wires
- Tactile switch

pressed more than once before the timing ends, it executes only once. This prevents the motor from restarting suddenly.

The circuit is isolated by an optocoupler (IC1) so that any disturbance in the motor power supply doesn't affect the Arduino Uno board. The freewheeling diode (D1)